FIRM: FHIR-Integrated Interviewing with Replaceable Modules for LLM-Assisted Pre-Consultation

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Abstract

Purpose: Large language model (LLM) assistants for pre-consultation must behave predictably and exchange data with electronic health records (EHRs). We present ELVINA (EHR-Linked Virtual Interviewer with Natural-language Assistance), a pre-consultation system built on FIRM (FHIR-Integrated Interviewing with Replaceable Modules), in which physician-authored questionnaires, not the LLM, select questions and results are written back as Taiwan Core (TW Core) FHIR resources, and we verify it against explicit requirements.

Methods: Questionnaires are encoded as FHIR Questionnaire. LLMs only extract evidence-grounded chief-complaint findings, classify EHR history into

questionnaire options, and draft reports; one transaction writes QuestionnaireResponse, Observation, Composition, and Provenance. Ten requirements (R0–R9) were checked with synthetic Synthea patients and 30 author-drafted scenarios answered by an LLM-simulated patient: interoperability, end-to-end runs, configuration-only model replacement, and local load. Clinician confirmation was simulated.

Results: Requirements R0, R1, R2, R3, R4, R6 passed and R5 failed. Question selection (R0) was Pass, and structural/profile checks of 1,348 written resources (R1, without terminology resolution) were Pass. After EHR-history classification was added, written-FHIR and runtime question paths agreed in 100 of 100 comparable sessions (R3, Pass). Replacing the commercial model by configuration alone was Pass; a local open-weight model was Fail. Red-flag rule coverage (R5) was Fail; candidate rules await physician approval. National-scale capacity was Not verifiable and exchange-hub integration Blocked.

Conclusions: A rule-governed pre-consultation system with FHIR write-back can be verified requirement by requirement. These provisional, synthetic software-verification results do not establish clinical safety, diagnostic performance, or deployment readiness.

Keywords: FHIR, Pre-consultation, Conversational AI, Clinical Documentation

1 Introduction

Low-acuity visits place a growing burden on unscheduled care worldwide.Pines et al. (2011) In Taiwan, despite low-cost access under National Health Insurance, its emergency departments remain chronically crowded, and about one-third of visits are non-emergent.Huang et al. (2019) This crowding prolongs waits, worsens outcomes, and drives clinician burnout.Morley et al. (2018); Ghaseminejad et al. (2025)

Large language models (LLMs) now support numerous clinical tasks, including ambient scribing, EHR documentation, and clinical decision support.Nair et al. (2026) Agentic systems extend these capabilities through planning, tool use, and multi-step action, enabling closer integration with health-system workflows and conversational diagnosis.Tu et al. (2025) However, although various benchmarks now evaluate LLMs on realistic tasks and interactive encounters, most agentic systems remain confined to simulated settings.Liu et al. (2026)

Here, we present ELVINA, an LLM-based agent for adults presenting in Taiwanese urgent care centers. ELVINA conducts structured history-taking by text or voice, drafts a clinician-ready note, and supports decision-making with a ranked differential and not-to-miss conditions, alongside suggested examinations and imaging. To limit hallucination, the LLM only extracts clinical facts linked to the patient's own words, while safety screening, question selection, and diagnostic ranking follow auditable rules. ELVINA connects to EHRs via SMART on FHIR and was evaluated on benchmark.

2 Methods

2.1 Scope and implemented architecture

We verified ELVINA, FIRM's reference implementation, using synthetic data. Outcomes concern interchange, interview-to-write-back completion, model replacement, and local concurrency; clinical accuracy and note quality were outside scope.

ELVINA combines Vue patient and clinician interfaces, a FastAPI backend, local retrieval, and a HAPI FHIR R4 sandbox (Fig. 1) (Vue.js 2026; Ramírez 2026; HAPI FHIR Project 2026; HL7 International 2019b). Invitations yield scoped session cookies; the clinician launcher binds patient and encounter identifiers. The separate SMART surface uses another authorization flow (Mandel et al. 2016; HL7 International 2024). Verification exercised text HTTP APIs, using isolated databases and an explicit loopback development authorization bypass.

In Fig. 2, fixed questionnaire rules select the next enabled question and skip prefilled fields. Models classify prefill and extract chief complaints, indirectly affecting rule inputs, but do not select the next question. Keywords choose the questionnaire first; only no match permits evidence-grounded model routing. Failed chief assessment triggers handoff. Retrieval supports report drafts (Lewis et al. 2020), without establishing diagnoses or orders. The alternative engine making per-turn model calls was not evaluated.

2.2 Conservative EHR prefill normalization

UCC, launcher, and sandbox ingress deterministically number chronic-history items. The configured report-extraction adapter returns only indices and fixed categories: diabetes, chronic kidney disease, hyperlipidemia, cirrhosis, autoimmune disease, cancer, other chronic, exclude, or unclear. Listed categories become questionnaire options; other chronic retains original text. Any unclear item, provider failure, timeout, or invalid/incomplete indices leaves both history fields unfilled for patient answers. Otherwise, cancer/autoimmune/other chronic items also prefill `chronic_detail`; enabled but prefilled questions are skipped. No recognized condition leaves `chronic` unfilled, without inferring "none." Original history, excluded items, classifications, model/prompt provenance, and bounded usage remain available for physician review. Oversized inputs are explicitly rejected; failures never echo history. Normalization sends history to the report provider, defaults off, and was enabled for these runs.

2.3 Questionnaire and write-back contracts

The encoder maps fields to `linkId`s, options to codes, and conditions to `enableWhen`. Questionnaire defines the form; QuestionnaireResponse records answers (HL7 International 2019a, 2022). Source JSON drives execution. Route selection pins the source-digest canonical version; write-back rejects drift and tags unpinned legacy resolution. R3 checks representation agreement.

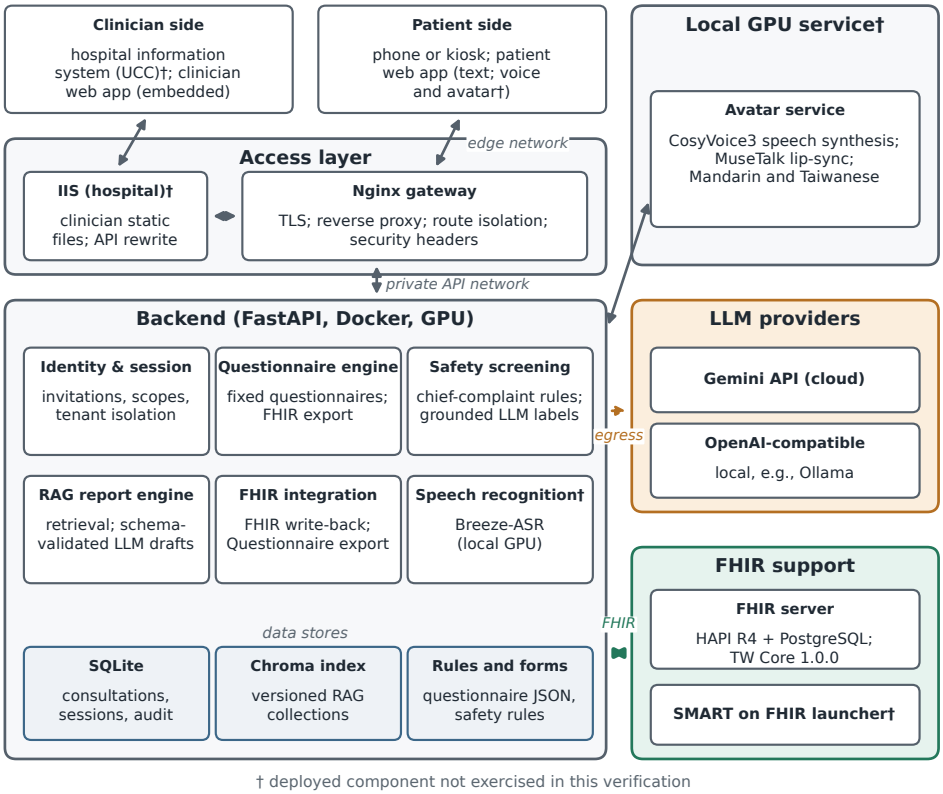


Fig. 1 ELVINA system architecture. Clinician traffic passes through the hospital IIS and the Nginx gateway; patient traffic reaches the gateway directly. The FastAPI backend and the local GPU avatar service share a private API network; only the backend has egress to model providers, and FHIR services are external to the backend. Daggers mark deployed components not exercised in this verification.

After simulated clinician confirmation, opt-in structured write-back creates Table 1's transaction; Composition-only remains default. Conditional creation supports retries. Patient/encounter/institution/issuer checks precede persistence. Differential diagnoses, orders, and safety Flags are not created.

Absent-EHR placeholders are omitted. Out-of-grammar text uses coded "other" with nested original text, avoiding the validator's open-choice exception. Normalized answers retain `ehr-prefill` source and add normalization/prompt-version extensions. A runtime answer that FHIR would drop causes non-retryable rejection; unanswered branches do not block writing.

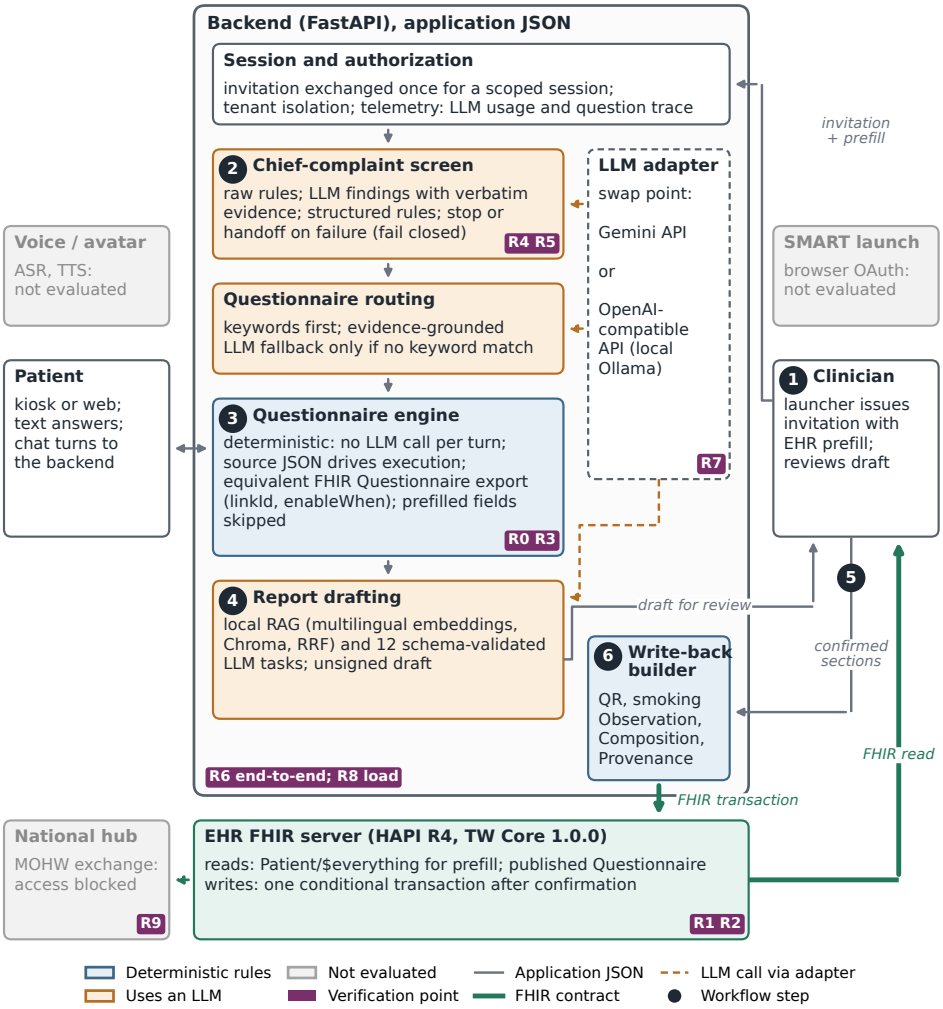


Fig. 2 FIRM workflow as implemented in ELVINA, with contracts and verification points. Workflow steps: (1) the clinician launcher reads EHR history and issues an invitation with prefill; (2) chief-complaint screening and questionnaire routing; (3) deterministic questionnaire turns; (4) report drafting; (5) clinician review and confirmation; (6) one FHIR write-back transaction. Source JSON drives the questionnaire engine and an equivalent FHIR Questionnaire is exported. Internal APIs use application JSON; FHIR defines the EHR boundary. Model-dependent components reach models only through the replaceable LLM adapter. Tags mark where requirements R0–R9 (Table 2) are checked; grey components were not evaluated.

Table 1 Write-back resources and assertion boundaries.

Resource	Content and boundary
QuestionnaireResponse	Ordered answers, source markers, and the pinned canonical reference.
Observation	Smoking status only for an explicit mapped patient selection; EHR text is not promoted to a new patient assertion.
Composition	Preliminary narrative, confirmed sections, and links to structured answers.
Provenance	Patient informant, AI assembly, and simulated clinician-review roles; no actual clinical approval.

2.4 Synthetic inputs and model disclosure

Synthea generated the adult source cohort (Walonoski et al. 2018; The MITRE Corporation 2026). Conversion applies fictional Taiwanese identifiers/addresses and TW Core profiles (Ministry of Health and Welfare, Taiwan 2025), retaining seeds, mappings, dropped elements, and hashes. The 30 author-drafted chest-pain, headache, and abdominal-pain scenarios include expected stops and known gaps; the planned benchmark was unavailable. Separate Synthea modules supply consistent backgrounds without manually rewriting generated records. Input validation is separate from output validation.

Background audits are mechanical; approval requires two independent physicians reviewing current script hashes, bindings, and paths. 0 scripts qualify. Online Resource 1 provides prompts/settings. Commercial calls use Gemini's API (Google 2026); Gemma/Qwen use Ollama (Ollama 2026; Gemma Team et al. 2025; Qwen Team 2025). Requested/transmitted settings are distinguished: newer Gemini configurations omit temperature. Unrecorded settings are not inferred.

2.5 Requirements and acceptance criteria

Table 2 maps requirements to generated verdicts. E1–E5 are verification activities. E1 zero errors, E3 zero code changes, and E4's request-error stopping threshold were prespecified. Requirement-defined all-match/all-pass criteria for R2–R6, R7 behavioral agreement, and the investigator's R0 fixed-selection requirement were applied post hoc on 2026-10-14. These are software criteria, not prospective clinical thresholds; missing evidence is indeterminate.

2.6 Verification activities

R0 audits saved calls for prefill normalization, chief extraction (at most once per session), report generation, or RAG translation, and requires identical repeated paths per script. Static inspection identifies the pure next-question selector. Prefill classification can change skip decisions; cross-model paths are descriptive. Bounded, incomplete usage cannot establish a complete call history.

E1 uses the pinned HL7/TW Core validator and generated definitions (HL7 International 2026; Ministry of Health and Welfare, Taiwan 2025). R1 requires zero

Table 2 Requirements, criterion sources, and generated verdicts. R0 uses E2 with E3 descriptive checks; R1/R2 use E1; R3–R6 use E2; R7 uses E3; R8 uses E4; R9 concerns E5. R7 is reported separately for each replacement in Table 5.

ID	Requirement	Criterion source	Verdict
R0	Questionnaire rules select the next question	Principal investigator requirement; criteria applied post hoc	Pass
R1	TW Core structural/profile checks	Prespecified zero errors; no validator exceptions	Pass
R2	Write/read-back identity	Requirement-defined, applied post hoc	Pass
R3	Top-level path matches the FHIR Questionnaire	Requirement-defined, applied post hoc	Pass
R4	Extracted-fact traceability	Requirement-defined, applied post hoc	Pass
R5	Scripted safety-rule execution	Requirement-defined, applied post hoc	Fail
R6	End-to-end completion without handoff	Requirement-defined, applied post hoc	Pass
R7	Configuration-only substitution retaining default behaviour	Prespecified zero code lines; default-equivalence applied post hoc	Alternate: Pass Local: Fail
R8	National peak-load capacity	No national-demand criterion available	Not verifiable
R9	MOHW hub interoperability	Access prerequisite unavailable	Blocked

written-resource errors/exceptions across templated and E2 cohorts. Checks use `-tx n/a`, without terminology resolution. R2 requires substantive write/read-back identity after excluding specified server metadata/identifiers. Missing outcomes cannot pass; supplemental E1 is counted once (Online Resource 1, Sections S3 and S7).

E2 spans invitation through read-back. A distinct Qwen model plays the patient from scenario facts, following simulated-patient evaluation (Tu et al. 2025; Johri et al. 2025; Schmidgall et al. 2024). Invalid constrained answers receive explicit unknowns, never expected answers. R3 compares presented top-level questions with `enableWhen` paths derived from written QuestionnaireResponse answers, including EHR prefill. Recorded prefilled fields retain skip context for no-data items omitted from QR; nested `<linkId>.other` details are not questions. Runtime-answer and script-derived checks remain secondary. Expected chief stops are excluded; unexpected stops fail. R4 requires every persisted chief fact to have traceable utterance evidence. R5 requires all scripted red-flag and non-red-flag expectations, measuring rule execution rather than clinical sensitivity/specificity. R6 requires write-back and `completed` or `summary_ready`; expected stops and successful retries qualify, manual handoffs do not.

E3 retains executable fingerprints across model configurations. R7 requires zero code changes and default-equivalent R1–R6 verdicts and per-script completion, paths, and red flags; shared failures remain in the baseline. One-time adapter changes against `1b0eb1c` are documented separately.

E4 replays turns using fixed-latency mocks or real models on backend v0.19.0, one worker, stopping above 5% request errors. Session failure is distinct; this local rule does not define national capacity. E5 needs MOHW exchange-hub access. Frozen runs and source hashes generate results; repetitions are not independent clinical observations.

Table 3 E1 written-resource verification. Input checks are excluded; exceptions would prevent acceptance.

Cohort	Resource	n	With errors	Errors	Warnings
Templated cohort	Composition	200	0	0	200
Templated cohort	Observation	198	0	0	0
Templated cohort	Provenance	200	0	0	800
Templated cohort	QuestionnaireResponse	200	0	0	0
Scenario cohort	Composition	150	0	0	150
Scenario cohort	Observation	100	0	0	0
Scenario cohort	Provenance	150	0	0	600
Scenario cohort	QuestionnaireResponse	150	0	0	0

Table 4 E1 workflow and semantic round trip. Read uses all sessions; write uses attempted writes; read-back uses returned references; round trip uses all captured submissions, including those without read-back.

Cohort	Sessions	Read	Write	Read-back	Round-trip
Templated cohort	200	100.0% (200/200)	100.0% (200/200)	100.0% (798/798)	100.0% (798/798)
Scenario cohort	150	100.0% (150/150)	100.0% (150/150)	100.0% (550/550)	100.0% (550/550)

3 Results

3.1 R0: deterministic question selection

R0 is **Pass**. Among 1,460 saved calls, 0 tasks fall outside the allowed list; chief extraction peaks at 1 per session. 150 usage records are incomplete. Paths are identical in 30/30 scripts across 5 repetitions. E3 alternate Gemini matches default paths in 30/30; Gemma matches in 24/30. Routing records show 100 keyword selections, 0 extraction fallbacks, and 50 unknown sources; unknowns are not counted as keywords.

3.2 R1 and R2: written resources and read-back

R1 is **Pass** and R2 is **Pass**. Across 200 templated and 150 E2 sessions, 1,348 written resources have 0 structural/profile errors and 0 validator exceptions under `-tx n/a` (Tables 3 and 4). Of 350 QuestionnaireResponses, 0 errors and 0 unverified responses are recorded. Warnings comprise 1,400 Provenance terminology and 350 Composition narrative warnings. The pinned participant-type CodeSystem remains unavailable.

3.3 R3–R6: questionnaire, evidence, rules, and completion

R3 is **Pass**: written-FHIR paths match in 100/100 comparable sessions, versus 100/100 using runtime answer strings. 50 expected chief stops are excluded; 0 paths are unknown. Patient-entered controls agree in 45/45, whereas EHR-prefilled controls agree in 55/55 (Online Resource 1, Section S6.3). R4 is **Pass**, with traceability of

Table 5 E3 requirement verdicts by configuration. R7 compares each replacement with the default, including per-script completion, path, and red-flag outcomes; shared rule gaps do not themselves fail this relative check. Code lines exclude one-time adapter implementation.

Check	Default Gemini	Alternate Gemini	Local Gemma 3
R1	Pass	Pass	Pass
R2	Pass	Pass	Pass
R3	Pass	Pass	Fail
R4	Pass	Pass	Pass
R5	Fail	Fail	Fail
R6	Pass	Pass	Fail
Code lines	0	0	0
R7	Reference	Pass	Fail

100.0% (458/458). R5 is **Fail**; nonmatching scripts are AP-10, CP-09 (full denominators in Online Resource 1). R6 is **Pass**, with completion of 100.0% (150/150), including 50 expected chief stops. Simulator fidelity is reported separately.

3.4 Descriptive prefill evidence

Normalization supplies answers in 184/200 E1 and 90/150 E2 sessions. E2 has 0 unclear items and 0 failures; Gemma prefill has 1 timeout. Separate feasibility matching is 96.4% over 2308 provisionally labeled entries; truncated inputs include 15/105 guessed items (Online Resource 1).

3.5 R7: configuration-only replacement

R7 is **Pass** for alternate Gemini and **Fail** for local Gemma (Table 5). The comparison checks executable fingerprints and default-relative requirement/per-script outcomes. Relative agreement does not establish safety. Gemma differs on AP-03, AP-04, AP-07, CP-04, HA-02, HA-08. R3 is Pass/Pass for default/alternate configurations. Gemma's R3 is Fail (19/22), with 3 unexpected chief handoffs rather than coding divergences; completion is 27/30.

3.6 R8 and R9: unavailable acceptance evidence

R8 is **Not verifiable** without national-demand evidence (`TODO_PEAK_DEMAND`); local stopping rules cannot substitute. R9 is **Blocked** without exchange-hub access. Local load, timing, and bounded usage/cost observations are descriptive (Online Resource 1).

4 Discussion

4.1 Principal findings

ELVINA keeps question selection under physician-authored rules and confines LLMs to bounded tasks: chief-complaint extraction with verbatim evidence, classification of

EHR history into questionnaire options, and draft reports. The verification shows that this division, which FIRM prescribes, can be checked requirement by requirement. Question selection (R0) was Pass; written-resource conformance and read-back (R1, R2) were Pass and Pass; written and runtime question paths (R3) were Pass; fact traceability (R4) and end-to-end completion (R6) were Pass and Pass. Replacing the commercial model through configuration alone was Pass, supporting the claim that model-dependent components sit behind a replaceable adapter.

LLM-led diagnostic dialogue systems select their own questions (Tu et al. 2025), and evaluations that require models to elicit information through conversation report weaker performance than vignette-based testing (Johri et al. 2025; Schmidgall et al. 2024). ELVINA trades conversational flexibility for predictable, auditable question selection; that predictability is what made path-level verification possible. EHR-history classification does influence which history questions are skipped, but the next question is still chosen by the questionnaire rules. Whether fixed questionnaires elicit as much relevant history as free dialogue is a separate empirical question not addressed here.

4.2 Defects found by verification

The checks exposed defects that component tests had not revealed. First, the runtime evaluated free-text EHR history by substring while the written FHIR representation used coded answers, so an earlier verification round found written paths that disagreed with the questions actually asked. The remediation classifies EHR history entries into questionnaire options by index and leaves both history questions to the patient whenever any entry is unclear or classification fails; in the selected runs R3 is Pass against the unchanged written-resource criterion. Second, prefilled history was silently truncated at a fixed length; oversized input is now rejected explicitly. Third, chief-complaint findings for suspected pulmonary embolism and for abdominal pain radiating to the back with a known aneurysm were extracted correctly, but no rule combined them (R5, Fail: AP-10, CP-09). Candidate rules drafted from current guidance (Konstantinides et al. 2020; Wanhainen et al. 2024) await physician approval and new physician-written scenarios, and were not enabled. Finally, screening runs only at the chief complaint, so later questionnaire selections such as fainting do not raise an alert; simulated-patient answers that contradicted their scripts illustrated this gap.

Configuration-level replaceability did not guarantee behavioural equivalence. The local open-weight model was Fail for R7 (AP-03, AP-04, AP-07, CP-04, HA-02, HA-08), through chief-stage handoffs after timeouts and missed expected stops; its R3 failures also reflect those handoffs rather than path disagreement. Each model replacement therefore needs re-verification against the same requirements. National capacity (R8) needs an authoritative demand target, and exchange-hub interoperability (R9) needs authorized test access.

4.3 Limitations and clinical review needs

Verdicts remain provisional pending physician review of the scenarios and independent audits. Requirement-defined criteria were applied post hoc, and clinician confirmation

was simulated. The LLM-simulated patient sometimes contradicted its script; these answers were recorded as patient-reported and limit scenario fidelity rather than the path and write-back checks. EHR-history classification is not clinical confirmation: the default model assigned some entries to listed options that a provisional reference does not support, such as prediabetes as diabetes and colon polyps as cancer (70 entries in the templated cohort), excluded some chronic conditions, and was not fully consistent across sessions (Online Resource 1, Section S6). These answers are written back as unconfirmed EHR prefill, and the original EHR text remains available to clinicians; the classifier was not modified for this study. Its reference labels lack physician review, and synthetic disease-name distributions do not establish accuracy on real EHR data. Structural/profile checks ran without terminology resolution. Synthetic demographics and sandbox interoperability do not establish clinical utility, production authorization, tenant isolation, browser SMART integration, or deployment readiness; terminology review, institutional governance, and human approval remain necessary.

5 Conclusion

ELVINA, built on the FIRM framework, keeps question selection under physician-authored rules and writes results back as TW Core FHIR resources, and it can be verified against explicit requirements. Requirements R0, R1, R2, R3, R4, R6 passed and R5 failed in this software verification; the failures identify concrete defects with documented remediations or pending clinical decisions. Replacing the commercial model by configuration was Pass, whereas a local open-weight model was Fail. Clinical validation, national-scale capacity (Not verifiable), and exchange-hub integration (Blocked) remain future work.

Supplementary information. Online Resource 1 documents methods, results, and limitations. Online Resource 2 supplies machine-readable synthetic materials, configurations, and provenance. Scenario approvals and the clinician screenshot remain pending.

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Declarations

Funding, competing interests, and contributions. Author statements and final approval are pending.

Ethics and consent. Only synthetic data were used. Authors must confirm the applicable institutional determination; no ethics approval is asserted.

Data availability. Online Resource 2 contains synthetic materials and selected results/provenance; full sessions and RAG source text are excluded. Medscape (Medscape 2026) redistribution rights require author confirmation (`TODO_AUTHOR`).

Code availability. Reproduction commands and source paths accompany Online Resource 2; the public repository location, archival identifier, and code/data licenses remain `TODO_AUTHOR`.

AI assistance. Claude Code and Codex assisted implementation, scenario drafting, orchestration, source checking, and manuscript revision. Model/tool disclosures and unrecorded versions appear in Online Resource 1, Section S5. Authors must confirm disclosure details, verify citations and content, and approve the text (`TODO_AUTHOR`); independent audits remain pending.

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